MATHS BRIDGE ACADEMY

ICSE Class X • Prelims – 1 • MATHEMATICS

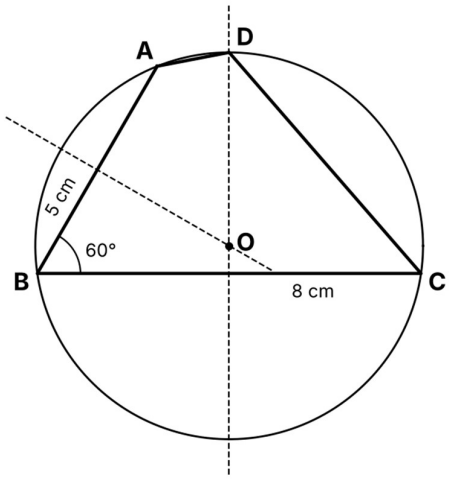
Maximum Marks: 80 Time: 3 hours

ANSWER KEY WITH COMPLETE SOLUTIONS

SECTION A (40 Marks)

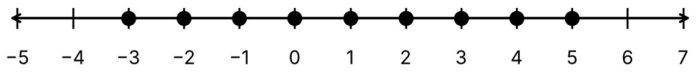
Question 1		
(i)	**(d) No real roots** $D = b^2 - 4ac = (-9)^2 - 4(2)(13) = 81 - 104 = -23 < 0$	**[15]**
(ii)	**(a) ₹ (24 × 800) + ₹ 1800** Maturity value = Total deposit + Interest = ₹ (24×800) + ₹ 1800 [2 years = 24 months]	
(iii)	**(c) $\frac{2}{5}$** Multiples of 3 or 7 $= \{3, 6, 9, 12, 15, 18, 7, 14\}$ $\therefore P = \frac{8}{20} = \frac{2}{5}$	
(iv)	**(a) Assertion (A) is true but reason (R) is false.** Reflection in y-axis: $(x, y) \to (-x, y)$, so $P(-2, 7) \to (2, 7)$. Only the sign of the x-coordinate changes.	
(v)	**(c) Both (A) and (R) are true and (R) is the correct explanation of (A).** $\frac{AB^2}{DE^2} = \frac{A(\triangle ABC)}{A(\triangle DEF)} = \frac{36}{49} \to \frac{AB}{DE} = \frac{6}{7}$	
(vi)	**(d) $\begin{bmatrix} 8 & 9 \\ 7 & 7 \end{bmatrix}$** $A + 5I = \begin{bmatrix} 3 & 9 \\ 7 & 2 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 8 & 9 \\ 7 & 7 \end{bmatrix}$	
(vii)	**(b) 45°** $\angle OPQ = 90°$ (radius ⊥ tangent). $\triangle OPQ$ isosceles $\Rightarrow OP = PQ$ $\therefore \angle OQP = \angle POQ = \frac{90°}{2} = 45°$	
(viii)	**(c) p = r** $f(2) = 0 \to 4p + 10 + r = 0$ and $f\left(\frac{1}{2}\right) = 0 \to \frac{p}{4} + \frac{5}{2} + r = 0 \to p + 10 + 4r = 0$ Subtracting: $3p - 3r = 0 \to p = r$	
(ix)	**(d) ₹ 4850** Money required = $100 \times$ ₹ 48.50 = ₹ 4850	
(x)	**(c) -1** $d = \frac{1-p}{p} - \frac{1}{p} = \frac{-p}{p} = -1$	
(xi)	**(b) $\sqrt{2} - 1$** Dividing by $\cos\theta$: $\tan\theta + 1 = \sqrt{2} \to \tan\theta = \sqrt{2} - 1$	

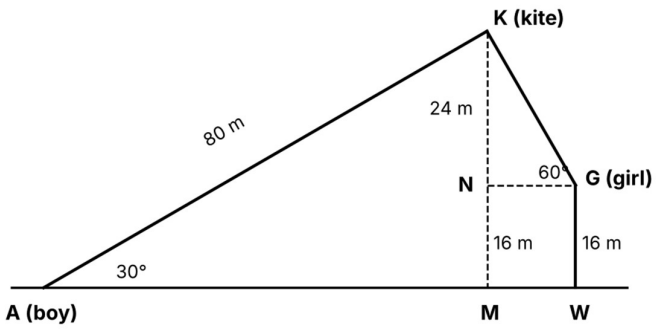
(xii)	**(c)** $\left(\frac{8}{3}\right)^{\frac{1}{3}}$ Half volume $= 11a^3 = 7 \times \frac{4}{3}\pi r^3 \rightarrow \frac{a^3}{r^3} = \frac{28}{33} \times \frac{22}{7} = \frac{8}{3} \rightarrow \frac{a}{r} = \left(\frac{8}{3}\right)^{\frac{1}{3}}$	
(xiii)	**(b) 22** Median through A passes through mid-point of $BC = \left(\frac{4+8}{2}, \frac{3-1}{2}\right) = (6, 1)$ $3(6) + 4(1) - k = 0 \rightarrow k = 22$	
(xiv)	**(c) Statement 1 is true and statement 2 is false.** $7x + 3 < 5x + 9 \rightarrow 2x < 6 \rightarrow x < 3$ (Statement 1 true). The graph shows $-2 \leq x < 4$ (Statement 2 false).	
(xv)	**(a) Mean** Median and quartiles are found from an ogive, mode from a histogram.	
Question 2		
(i)	$5x^2 - 3x - 4 = 0$. Comparing with $ax^2 + bx + c = 0$: $a = 5,\ b = -3,\ c = -4$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(5)(-4)}}{2 \times 5}$ $x = \frac{3 \pm \sqrt{9 + 80}}{10} = \frac{3 \pm \sqrt{89}}{10} = \frac{3 \pm 9.434}{10}$ $x = \frac{12.434}{10}$ or $x = \frac{-6.434}{10}$ $x = 1.2434$ or $x = -0.6434$ $\therefore$ **x = 1.24** or **x = −0.643** (correct to 3 significant figures)	**[4]**
(ii)	$n = 2$ years $= 24$ months, $r = 6\%$ p.a., $I = ₹\,1200$. Let monthly instalment $= ₹\,P$ (i) $I = P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100}$ $1200 = P \times \frac{24 \times 25}{2 \times 12} \times \frac{6}{100}$ $1200 = P \times \frac{3}{2} \rightarrow P = \frac{1200 \times 2}{3} = 800$ $\therefore$ **Monthly instalment = ₹ 800** (ii) Maturity value $= P \times n + I = 800 \times 24 + 1200 = 19200 + 1200$ $\therefore$ **Amount at maturity = ₹ 20400**	**[4]**
(iii)	Let $P(4, b)$ divide the join of $A(6, -2)$ and $B(-3, 16)$ in the ratio $m : n$. By section formula, $x = \frac{mx_2 + nx_1}{m+n} \rightarrow 4 = \frac{m(-3) + n(6)}{m+n}$ $4m + 4n = -3m + 6n \rightarrow 7m = 2n \rightarrow \frac{m}{n} = \frac{2}{7}$ $\therefore$ **Ratio = 2 : 7** $b = \frac{my_2 + ny_1}{m+n} = \frac{2(16) + 7(-2)}{2+7} = \frac{32 - 14}{9} = \frac{18}{9}$ $\therefore$ **b = 2**	**[4]**
Question 3		
(i)	**Steps of construction:**	**[4]**

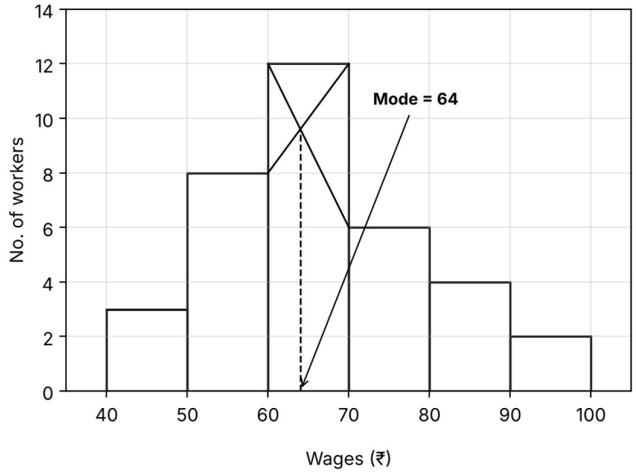
	1. Draw $BC = 8$ cm. At B, construct $\angle CBX = 60°$ and cut off $BA = 5$ cm on BX. 2. Draw the perpendicular bisectors of AB and BC; let them meet at O. 3. With centre O and radius OB, draw the circumcircle through A, B and C. 4. D is equidistant from B and C, so D lies on the perpendicular bisector of BC. Let this bisector cut the circle at D (on the opposite side of AC from B). 5. Join AD and CD. $ABCD$ is the required cyclic quadrilateral. 	
(ii)	From the graph: $A(6,0),\ B(0,-3),\ C(-2,0),\ D(0,4)$ $\therefore OA = 6,\ OB = 3,\ OC = 2,\ OD = 4$ units In $\triangle AOB$ and $\triangle DOC$: $\angle AOB = \angle DOC = 90°$ (axes are perpendicular) $\frac{OA}{OD} = \frac{6}{4} = \frac{3}{2}$ and $\frac{OB}{OC} = \frac{3}{2}$ $\therefore \frac{OA}{OD} = \frac{OB}{OC}$ $\therefore \triangle AOB \sim \triangle DOC$ (SAS similarity) **Hence proved.** $\frac{A(\triangle AOB)}{A(\triangle DOC)} = \frac{OA^2}{OD^2} = \left(\frac{3}{2}\right)^2 = \mathbf{\frac{9}{4}}$ (ratio of areas of similar $\triangle$s = ratio of squares of corresponding sides)	**[4]**
(iii)	Let the digits (hundreds, tens, units) be $a-d,\ a,\ a+d$. Sum of digits: $(a-d)+a+(a+d) = 15 \rightarrow 3a = 15 \rightarrow a = 5$ Original number $= 100(5-d)+10(5)+(5+d) = 555-99d$ Reversed number $= 100(5+d)+10(5)+(5-d) = 555+99d$ Given: $(555+99d)-(555-99d) = 792$ $198d = 792 \rightarrow d = 4$ Digits are $1, 5, 9$ $\therefore$ **Original number = 159** [Check: $951-159 = 792$]	**[4]**

SECTION B (40 Marks)

Question 4		
(i)	$-2\frac{1}{6} \le \frac{x}{3} - 1\frac{1}{6} < \frac{5}{6},\ x \in I$ $-\frac{13}{6} \le \frac{x}{3} - \frac{7}{6} < \frac{5}{6}$	**[3]**

	Adding $\frac{7}{6}$ throughout: $-\frac{6}{6} \le \frac{x}{3} < \frac{12}{6} \rightarrow -1 \le \frac{x}{3} < 2$ Multiplying by 3: $-3 \le x < 6$ **Solution set** $= \{-3, -2, -1, 0, 1, 2, 3, 4, 5\}$  	
(ii)	(i) Order of $\begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix}$ is 2×2 and order of product is 2×1 $\therefore$ **Order of X is 2×1** (ii) Let $X = \begin{bmatrix} a \\ b \end{bmatrix}$ $\begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2a+b \\ -3a+4b \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ $2a + b = 7 \ \ldots(1)$ and $-3a + 4b = 6 \ \ldots(2)$ From (1): $b = 7 - 2a$. Substituting in (2): $-3a + 4(7 - 2a) = 6 \rightarrow -11a = -22 \rightarrow a = 2,\ b = 3$ $\therefore X = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$	**[3]**
(iii)	$LHS = \frac{\cot^2 A}{(\text{cosec } A + 1)^2}$ $= \frac{\text{cosec}^2 A - 1}{(\text{cosec } A + 1)^2}$ $[\because 1 + \cot^2 A = \text{cosec}^2 A]$ $= \frac{(\text{cosec } A - 1)(\text{cosec } A + 1)}{(\text{cosec } A + 1)^2} = \frac{\text{cosec } A - 1}{\text{cosec } A + 1}$ $= \frac{\frac{1}{\sin A} - 1}{\frac{1}{\sin A} + 1} = \frac{\frac{1 - \sin A}{\sin A}}{\frac{1 + \sin A}{\sin A}}$ $= \frac{1 - \sin A}{1 + \sin A} = RHS$ **Hence proved.**	**[4]**
Question 5		
(i)	$\frac{x^2 + x - 3}{x - 3} = \frac{4x^2 + 3x - 2}{3x - 2}$ Applying dividendo: $\frac{(x^2 + x - 3) - (x - 3)}{x - 3} = \frac{(4x^2 + 3x - 2) - (3x - 2)}{3x - 2}$ $\frac{x^2}{x - 3} = \frac{4x^2}{3x - 2}$ $x^2(3x - 2) = 4x^2(x - 3) \rightarrow x^2[(3x - 2) - 4(x - 3)] = 0$ $x^2(10 - x) = 0$ $\therefore$ **x = 0** or **x = 10**	**[3]**
(ii)	Let $f(x) = 3x^3 + 2x^2 - 19x + 6$ $f(2) = 3(2)^3 + 2(2)^2 - 19(2) + 6 = 24 + 8 - 38 + 6 = 0$ $\therefore (x - 2)$ is a factor of $f(x)$ (remainder $= 0$)	**[3]**

	$$\begin{array}{r} 3x^2 + 8x - 3 \\ x-2\,\overline{)\,3x^3+2x^2-19x+6} \\ \underline{3x^3-6x^2} \\ 8x^2-19x \\ \underline{8x^2-16x} \\ -3x+6 \\ \underline{-3x+6} \\ 0 \end{array}$$ $f(x) = (x-2)(3x^2+8x-3)$ $= (x-2)(3x^2+9x-x-3) = (x-2)[3x(x+3)-1(x+3)]$ $\therefore \mathbf{f(x) = (x-2)(x+3)(3x-1)}$	
(iii)	Let V be the vertex of the cone and x cm the radius of the sphere (centre O). Slant height $VT = \sqrt{3^2+4^2} = \sqrt{25} = 5$ cm $TB = TA = 3$ cm (tangents from an external point are equal) $\therefore VB = VT - TB = 5-3 = 2$ cm and $OV = AV - OA = (4-x)$ cm In $\triangle OBV$, $\angle OBV = 90°$ (radius $\perp$ tangent): $OB^2 + VB^2 = OV^2$ $x^2 + 2^2 = (4-x)^2 \rightarrow x^2+4 = 16-8x+x^2 \rightarrow 8x = 12 \rightarrow x = 1.5$ cm Volume of water that overflows = Volume of sphere $= \frac{4}{3}\pi(1.5)^3 = 4.5\pi$ cm^3 Volume of water in cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(3)^2(4) = 12\pi$ cm^3 **Fraction of water that overflows** $= \frac{4.5\pi}{12\pi} = \mathbf{\frac{3}{8}}$	**[4]**
Question 6		
(i)	(i) $(0,4)$ and $(0,-2)$ lie on the y-axis $\therefore L_1$ **is the y-axis:** $x = 0$ $(1,0)$ and $(-3,0)$ lie on the x-axis $\therefore L_2$ **is the x-axis:** $y = 0$ (ii) Reflection in y-axis: $(x,y) \rightarrow (-x,y)$ $\therefore P(-3,4) \rightarrow \mathbf{P'(3,4)}$ (iii) Reflection in x-axis: $(x,y) \rightarrow (x,-y)$ $\therefore P'(3,4) \rightarrow \mathbf{P''(3,-4)}$ (iv) $P(-3,4) \rightarrow P''(3,-4)$: **Reflection in the origin.**	**[5]**
(ii)	 Let A be the boy, K the kite and G the girl. $KM \perp$ ground, $GN \perp KM$, $GW = NM = 16$ m. In $\triangle AMK$: $\sin 30° = \frac{KM}{AK} \rightarrow \frac{1}{2} = \frac{KM}{80} \rightarrow KM = 40$ m $KN = KM - NM = 40 - 16 = 24$ m In $\triangle KNG$: $\sin 60° = \frac{KN}{KG} \rightarrow \frac{\sqrt{3}}{2} = \frac{24}{KG}$	**[5]**

	$KG = \dfrac{48}{\sqrt{3}} = \dfrac{48\sqrt{3}}{3} = 16\sqrt{3} = 16 \times 1.732 = 27.712$ m $\therefore$ **Distance between the girl and the kite = 27.71 m**	
Question 7		
(i)	**Steps:** 1. Draw the histogram with wages on the x-axis and number of workers on the y-axis. 2. In the highest rectangle (60–70), join its upper corners to the upper corners of the adjacent rectangles diagonally. 3. From the point of intersection, drop a perpendicular on the x-axis. It meets the axis at 64.  $\therefore$ **Mode = ₹ 64**	**[3]**
(ii)	In a kite, the diagonals are perpendicular and the longer diagonal AC bisects the shorter diagonal BD. Mid-point of BD: $M = \left(\dfrac{7+9}{2}, \dfrac{-1+(-5)}{2}\right) = (8, -3)$ Slope of $BD = \dfrac{-5-(-1)}{9-7} = \dfrac{-4}{2} = -2$ $\therefore$ Slope of $AC = \dfrac{1}{2}$ $(m_1 m_2 = -1)$ Equation of AC: $y - y_1 = m(x - x_1) \rightarrow y + 3 = \dfrac{1}{2}(x - 8)$ $2y + 6 = x - 8$ $\therefore$ **x − 2y − 14 = 0**	**[3]**
(iii)	Both articles cost more than ₹ 2000, so 5% discount applies to each. **Article A:** Discount = 5% of ₹ 4000 = ₹ 200 $\therefore$ Discounted price = ₹ 3800 GST = 18% of ₹ 3800 $= \dfrac{18}{100} \times 3800 =$ ₹ 684 $\therefore$ Amount = 3800 + 684 = ₹ 4484 **Article B:** Discount = 5% of ₹ 2400 = ₹ 120 $\therefore$ Discounted price = ₹ 2280 GST = 5% of ₹ 2280 $= \dfrac{5}{100} \times 2280 =$ ₹ 114 $\therefore$ Amount = 2280 + 114 = ₹ 2394 Total bill = ₹ 4484 + ₹ 2394 = ₹ 6878 $\therefore$ **Extra amount paid** = ₹ 7000 − ₹ 6878 = **₹ 122**	**[4]**
Question 8		
(i)	(i) Number of shares $= \dfrac{\text{Investment}}{\text{M.V.}} = \dfrac{8000}{80} = 100$ Dividend for first year = 7% of (₹ 100 × 100) $= \dfrac{7}{100} \times 10000 =$ **₹ 700**	**[3]**

(ii) Sale proceeds $= 100 \times ₹\,75 = ₹\,7500$

Amount invested in new shares $= ₹\,7500 + ₹\,700 = ₹\,8200$

Number of new shares $= \dfrac{8200}{41} = 200$

Annual income in second year $= 18\%$ of $(₹\,25 \times 200) = \dfrac{18}{100} \times 5000 = \mathbf{₹\,900}$

(iii) Increase in return $= ₹\,900 - ₹\,700 = ₹\,200$

Percentage increase on original investment $= \dfrac{200}{8000} \times 100 = \mathbf{2.5}\%$

(ii) **[3]**

Let assumed mean $A = 67.5$, class width $i = 5$, $\quad t = \dfrac{x - A}{i}$

Class	**Mid-value** x	f	$t = \frac{x-67.5}{5}$	ft
50–55	52.5	5	-3	-15
55–60	57.5	20	-2	-40
60–65	62.5	10	-1	-10
65–70	67.5	10	0	0
70–75	72.5	9	1	9
75–80	77.5	6	2	12
80–85	82.5	12	3	36
85–90	87.5	8	4	32
		$\sum f = 80$		$\sum ft = 24$

Mean $= A + \dfrac{\sum ft}{\sum f} \times i = 67.5 + \dfrac{24}{80} \times 5 = 67.5 + 1.5$

$\therefore$ **Mean marks = 69**

(iii) **[4]**

Five years ago, let the son's age be x years. Then the woman's age was x^2 years.

Present ages: son $= (x + 5)$ years, woman $= (x^2 + 5)$ years

Four years hence: son $= (x + 9)$ years, woman $= (x^2 + 9)$ years

Given: $\quad x^2 + 9 = 3(x + 9)$

$x^2 - 3x - 18 = 0 \rightarrow x^2 - 6x + 3x - 18 = 0 \rightarrow (x - 6)(x + 3) = 0$

$x = 6 \quad$ or $\quad x = -3 \quad$ (age cannot be negative, so $x = -3$ is rejected)

(i) **Age of son five years ago = 6 years**

(ii) **Present age of the woman** $= 6^2 + 5 =$ **41 years**

Question 9

(i) **[3]**

Cone: $r = \dfrac{7}{2}$ cm, $h = 3$ cm. Let radius of sphere $= R$ cm.

Volume of sphere $= 126\times$ Volume of one cone

$\dfrac{4}{3}\pi R^3 = 126 \times \dfrac{1}{3}\pi \left(\dfrac{7}{2}\right)^2 (3)$

$R^3 = \dfrac{126 \times 49 \times 3}{4 \times 4} = \dfrac{18522}{16} = \dfrac{9261}{8} \rightarrow R = \dfrac{21}{2} = 10.5$ cm

$\therefore$ **Diameter of sphere = 21 cm**

Surface area $= 4\pi R^2 = 4 \times \dfrac{22}{7} \times \dfrac{21}{2} \times \dfrac{21}{2} =$ **1386 cm^2**

(ii) **[3]**

Sample space $S = \{1, 2, 3, -1, -2, -3\}, \quad n(S) = 6$

	(i) Positive integers $= \{1, 2, 3\}$ $\therefore P(\text{positive integer}) = \frac{3}{6} = \mathbf{\frac{1}{2}}$ (ii) Integers greater than $-3 = \{1, 2, 3, -1, -2\}$ $\therefore P = \mathbf{\frac{5}{6}}$ (iii) Smallest integer $= \{-3\}$ $\therefore P(\text{smallest integer}) = \mathbf{\frac{1}{6}}$	
(iii)	(i) $\angle BCD + \angle DAB = 180°$ (opposite ∠s of a cyclic quadrilateral are supplementary) $\angle BCD = 180° - 55° = \mathbf{125°}$ (ii) $\angle BOD = 2 \times \angle DAB = 2 \times 55° = \mathbf{110°}$ (∠ at centre $= 2 \times$ ∠ at circumference) (iii) In $\triangle OBD$, $OB = OD$ (radii) $\therefore \angle ODB = \angle OBD = \frac{180° - 110°}{2} = \mathbf{35°}$ (iv) $\angle DBC = \angle ODB = 35°$ (alternate ∠s, $OD \parallel BC$) In $\triangle BCD$: $\angle BDC = 180° - (125° + 35°) = \mathbf{20°}$	**[4]**
Question 10		
(i)	Let first term $= a$ and common ratio $= r$. $T_n = ar^{n-1}$ $T_5 = ar^4 = 48 \ldots (1)$ and $T_8 = ar^7 = 384 \ldots (2)$ (2) ÷ (1): $\frac{ar^7}{ar^4} = \frac{384}{48} \rightarrow r^3 = 8 \rightarrow r = 2$ From (1): $a \times 2^4 = 48 \rightarrow a = 3$ $T_{12} = ar^{11} = 3 \times 2^{11} = 3 \times 2048$ $\therefore \mathbf{T_{12} = 6144}$	**[3]**
(ii)	<table><tr><th>Monthly salary (₹)</th><th>c.f. (from graph)</th><th>No. of workers f</th></tr><tr><td>0–2000</td><td>10</td><td>10</td></tr><tr><td>2000–4000</td><td>20</td><td>10</td></tr><tr><td>4000–6000</td><td>35</td><td>15</td></tr><tr><td>6000–8000</td><td>55</td><td>20</td></tr><tr><td>8000–10000</td><td>70</td><td>15</td></tr><tr><td>10000–12000</td><td>75</td><td>5</td></tr></table> (i) Last point on the ogive is $(12000, 75)$ $\therefore$ **Total number of workers = 75** (ii) $\frac{N}{2} = \frac{75}{2} = 37.5$, which lies between c.f. 35 and 55 $\therefore$ **Median class = ₹6000 – ₹8000** (iii) Workers with salary $\geq$ ₹6000 but $<$ ₹10000 = c.f. at 10000 − c.f. at 6000 $= 70 - 35 = \mathbf{35}$	**[3]**
(iii)	$\angle OAT = \angle OBT = 90°$ (radius $\perp$ tangent at the point of contact) In quadrilateral $OATB$: $\angle AOB = 360° - (90° + 90° + 70°) = \mathbf{110°}$ $\angle APB = \frac{1}{2}\angle AOB = \frac{1}{2} \times 110° = \mathbf{55°}$ (∠ at centre $= 2 \times$ ∠ at circumference) $\angle AQB = 180° - 55° = \mathbf{125°}$ (opp. ∠s of cyclic quad. $APBQ$ are supplementary) $TA = TB$ (tangents from an external point) $\therefore \angle TAB = \angle TBA$ $\angle TAB = \frac{180° - 70°}{2} = \mathbf{55°}$	**[4]**